Mechanical Work and Power

Fully Worked Answers

Numerical-format rule used throughout: Equation → Rearrange → Substitute SI values (without units) → Final answer with units.

Part A · Multiple Choice

1. C — Only the component parallel to displacement does work: $W = Fd\cos\theta$.
2. C — The support force is vertical while displacement is horizontal, so $\theta = 90°$.
3. B — Friction acts opposite the displacement.
4. C — Net work = 40 − 15 = 25 J, so $\Delta K = +25$ J.
5. B — $W_2/W_1 = \cos60°/\cos0° = 1/2$.
6. C — Work is the signed area under $F_\parallel$ versus displacement.
7. B — Area below the axis is negative work.
8. B — By the work-energy theorem, net work zero means $\Delta K = 0$.
9. C — $P_{inst} = Fv$ for a parallel force; doubling v doubles the instantaneous power.
10. B — With constant F and constant v, $P_{inst} = Fv$ is constant.
11. B — Pavg = $\Delta E/\Delta t = 2400/6 = 400$ W.
12. C — $P_{inst} = Fv\cos90° = 0$.
13. C — Conservative-force work depends only on initial and final configuration.
14. B — For friction of constant magnitude, dissipated energy magnitude is Ff times path length.
15. B — Power is the rate of energy change: slope of E versus t.
16. B — cos120° is negative, so the work is negative.
17. C — $W = (200)(3) + (-100)(2) = +400$ J.
18. C — Pinst = $Fv = (150)(4.0) = 600$ W.

FRQ 1 · Horizontal push, no friction

(a)

Equation: $W = F_{\parallel} d$

Rearrange: No rearrangement required.

Substitute: $W = (480)(3.5)$

Answer: $W = 1680\ \text{J}$

(b)

Equation: $\Sigma F = F_{net} = ma$

Rearrange: $a = F_{net}/m$

Substitute: $a = 480/360$

Answer: $a = 1.33\ \text{m s}^{-2}$

(c)

Equation: $v^2 = v_0^{\,2} + 2a\Delta x$

Rearrange: $v = \sqrt{(v_0^{\,2} + 2a\Delta x)}$

Substitute: $v = \sqrt{(1.0^2 + 2(1.333)(3.5))}$

Answer: $v = 3.21\ \text{m s}^{-1}$

(d)

Equation: $K = \tfrac{1}{2}mv^2$

Rearrange: No rearrangement required.

Substitute: $K_i = \tfrac{1}{2}(360)(1.0^2)$

Answer: $K_i = 180\ \text{J}$

(e)

Equation: $K = \tfrac{1}{2}mv^2$

Rearrange: No rearrangement required.

Substitute: $K_f = \tfrac{1}{2}(360)(3.2146^2)$

Answer: $K_f = 1860\ \text{J}$

(f)

Equation: $\Delta K = K_f - K_i$

Rearrange: No rearrangement required.

Substitute: $\Delta K = 1860 - 180$

Answer: $\Delta K = 1680\ \text{J}$

(g) First find the time interval.

Equation: $v = v_0 + at$

Rearrange: $t = (v - v_0)/a$

Substitute: $t = (3.2146 - 1.0)/1.333$

Answer: $t = 1.66\ \text{s}$

Equation: $P_{avg} = W/\Delta t$

Rearrange: No rearrangement required.

Substitute: $P_{avg} = 1680/1.6609$

Answer: $P_{avg} = 1.01 \times 10^3\ \text{W}$

(h) No. $P_{inst} = F_{\parallel} v$. The force is constant but the box speeds up, so the instantaneous power increases as the box moves.

FRQ 2 · Horizontal push with friction

(a)

Equation: $W = F_{\parallel} d$

Rearrange: No rearrangement required.

Substitute: $W = (480)(3.5)$

Answer: $W = 1680\ \text{J}$

(b) Constant speed means $a = 0$, so the net horizontal force is zero and $F_f = 480\ \text{N}$.

Equation: $F_f = \mu_k F_n$

Rearrange: $\mu_k = F_f/F_n = F_f/(mg)$

Substitute: $\mu_k = 480/((360)(9.8))$

Answer: $\mu_k = 0.136$

The person transfers energy into the box-floor system by doing positive work; friction transfers mechanical energy into internal/thermal energy. The box's kinetic energy is unchanged.

(c)

Equation: $P_{avg} = W/\Delta t$

Rearrange: Using $\Delta t = d/v$, $P_{avg} = Fv$

Substitute: $P_{avg} = (480)(1.0)$

Answer: $P_{avg} = 480\ \text{W}$

Because both F and v are constant, $P_{inst} = Fv$ is also constant at 480 W.

(d)

Equation: $W = F_{\parallel} d$

Rearrange: No rearrangement required.

Substitute: $W = (540)(3.5)$

Answer: $W = 1890\ \text{J}$

(e)

Equation: $W_f = F_f d \cos 180°$

Rearrange: No rearrangement required.

Substitute: $W_f = (480)(3.5)(-1)$

Answer: $W_f = -1680\ \text{J}$

(f) Net work is the sum of the applied-force work and friction work.

Equation: $\Delta K = \Sigma W_i$

Rearrange: No rearrangement required.

Substitute: $\Delta K = 1890 - 1680$

Answer: $\Delta K = 210\ \text{J}$

Equation: $K_f = K_i + \Delta K$

Rearrange: No rearrangement required.

Substitute: $K_f = \frac{1}{2}(360)(1.0^2) + 210$

Answer: $K_f = 390\ \text{J}$

Equation: $K = \frac{1}{2}mv^2$

Rearrange: $v = \sqrt{(2K/m)}$

Substitute: $v = \sqrt{((2)(390)/360)}$

Answer: $v = 1.47\ \text{m s}^{-1}$

FRQ 3 · Carrying a box

Equation: $W = Fd\cos\theta$

Rearrange: No rearrangement required.

Substitute: $W = F(16)\cos 90°$

Answer: $W = 0$ J

Assumptions: the box remains at constant height, there is no horizontal force from the person on the box, and the support force is vertical. The person may expend chemical energy biologically, but the upward support force does zero mechanical work on the box during the horizontal displacement.

FRQ 4 · Child and trolley

(a)

Equation: $W = Fd\cos\theta$

Rearrange: No rearrangement required.

Substitute: $W = (56)(6.0)\cos 28°$

Answer: $W = 297$ J

(b) Assume the trolley starts from rest and friction/air resistance are negligible. Then the father's work is the net work.

Equation: $\Delta K = \Sigma W_i$

Rearrange: $K_f = W$ because $K_i = 0$

Substitute: $K_f = 296.7$

Answer: $K_f = 297$ J

Equation: $K = \frac{1}{2}mv^2$

Rearrange: $v = \sqrt{(2K/m)}$

Substitute: $v = \sqrt{((2)(296.7)/42)}$

Answer: $v = 3.76\ \text{m s}^{-1}$

FRQ 5 · Variable net force

(a) Work is the area under the F_{net}-versus-displacement graph.

Equation: W_{net} = area under F_{net} vs x

Rearrange: Add two triangles and one rectangle.

Substitute: $W_{net} = ½(1.0)(400) + (2.0)(400) + ½(0.50)(400)$

Answer: $W_{net} = 1100$ J

(b)

Equation: $\Delta K = W_{net}$

Rearrange: $K_f = K_i + W_{net}$

Substitute: $K_f = ½(360)(1.0^2) + 1100$

Answer: $K_f = 1280$ J

Equation: $K = ½mv^2$

Rearrange: $v = \sqrt{(2K/m)}$

Substitute: $v = \sqrt{((2)(1280)/360)}$

Answer: $v = 2.67$ m s^{-1}

(c) The graph gives the net force, so it determines net work but does not, by itself, determine the work done by the person. If friction acts, the person's applied force and positive work would be larger than the net force/work. The mechanical energy K + U of the box-Earth system is not conserved here: U_g is unchanged while K increases by 1100 J, so energy is transferred into the selected system by external work.

FRQ 6 · Power while pulling at an angle

(a)

Equation: $W = Fd\cos\theta$

Rearrange: First find $d = v\Delta t$.

Substitute: $W = (90)((1.8)(12))\cos35°$

Answer: $W = 1592$ J

(b)

Equation: $P_{avg} = W/\Delta t$

Rearrange: No rearrangement required.

Substitute: $P_{avg} = 1592.4/12$

Answer: $P_{avg} = 133$ W

(c)

Equation: $P_{inst} = Fv\cos\theta$

Rearrange: No rearrangement required.

Substitute: $P_{inst} = (90)(1.8)\cos35°$

Answer: $P_{inst} = 133$ W

It is constant because F, θ and v are all constant.

(d) Constant speed means zero net horizontal force.

Equation: $\Sigma F_x = 0$

Rearrange: $F_f = F\cos\theta$

Substitute: $F_f = (90)\cos35°$

Answer: $F_f = 73.7$ N

FRQ 7 · Experimental / graphical

(a) Plot the measured parallel force $F_{\parallel}$ against displacement x. The signed area under the graph between the chosen initial and final positions is the work done by the applied force.

(b) Use the motion sensor to obtain the cart's initial and final speeds. Calculate $K_i = \frac{1}{2}mv_i^2$ and $K_f = \frac{1}{2}mv_f^2$, then $\Delta K = K_f - K_i$. Compare ΔK with the net work. To test the theorem accurately, either measure/account for other forces (for example friction) or make them negligible.

(c)

Equation: $P_{inst} = F_{\parallel}v$

Rearrange: No rearrangement required.

Substitute: $P_{inst} = (3.6)(1.25)$

Answer: $P_{inst} = 4.5$ W

(d) The work-energy theorem uses the sum of the work done by all forces. If friction or another force also does work, the applied-force work alone is not equal to ΔK.

Audit note

The original worksheet questions have been preserved in substance. Two source-answer issues were corrected in these worked solutions: the variable-force problem gives $K_f = 1280$ J (not 1290 J), and the corresponding final speed is 2.67 m s^{-1}.